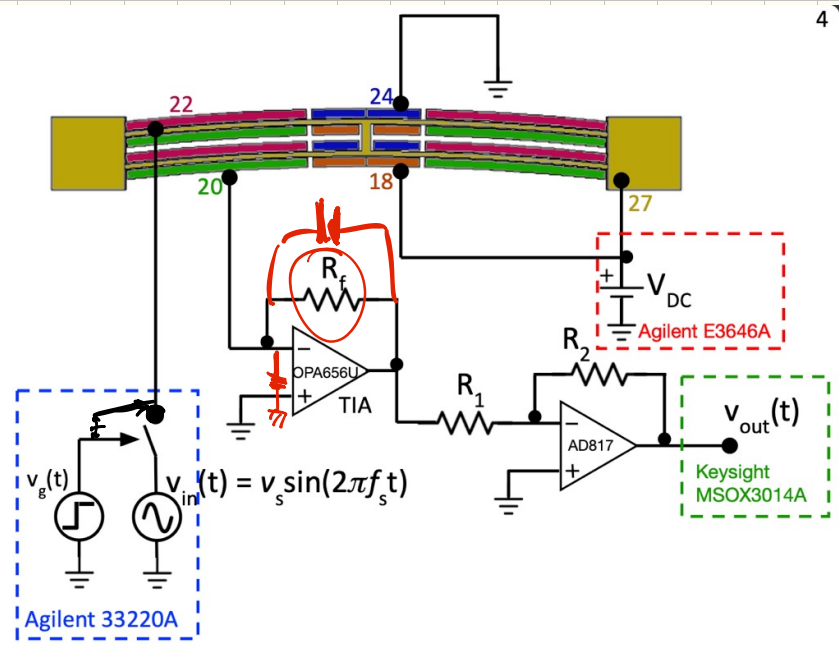




$$C_D = C_{22/27}$$

$$\Delta V_{22/27} = V_{OC} - V_{in}$$

$$F_e = \frac{1}{2} \frac{dC_D}{dx} \Delta V_{22/27}^2$$

$$= \frac{1}{2} Y_D (V_D - V_{in})^2$$

$$= \frac{1}{2} Y_D V_D^2 - Y_D V_D V_{in} + \frac{1}{2} Y_D V_{in}^2$$

piccolissimo
+ vibro doppio
frequenza

costante
fa stare legg. stanta MA
non fa vibrare

MEHS
→ attivato
→ molla
→ massa

Risonanza = 0 molla e massa
Si annullano!

$$F_e = F_A + \cancel{F_m} + \cancel{F_k}$$

$$F_e = F_A = \alpha U \Rightarrow U = \frac{F_e}{\alpha}$$

$$C_S = C_{20/27} \Rightarrow Q_S = C_S \cdot \Delta V_{20/27} = C_S \cdot V_{DC}$$

$$i_{out} = \frac{dQ_S}{dt} = V_{DC} \frac{dC_S}{dt} = V_{DC} \frac{dC_S}{dx} \frac{dx}{dt} = V_{DC} \cdot Y_S \cdot U = V_{DC} Y_S \frac{F_e}{\alpha}$$

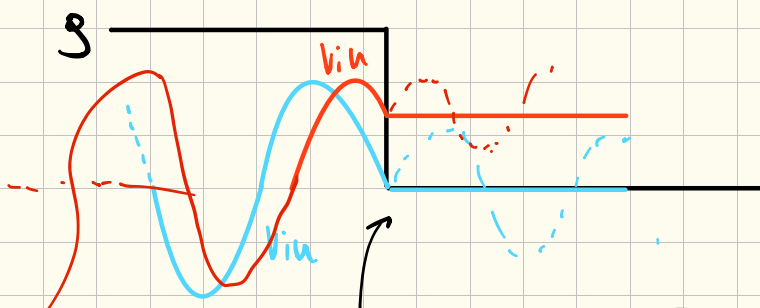
Quando stacco V_{in} :

$$U(t) = U_0 e^{-\frac{t}{\tau}} \sin(2\pi f_s t)$$

$$I_{out} = I_0 e^{-\frac{t}{\tau}} \sin(2\pi f_s t)$$

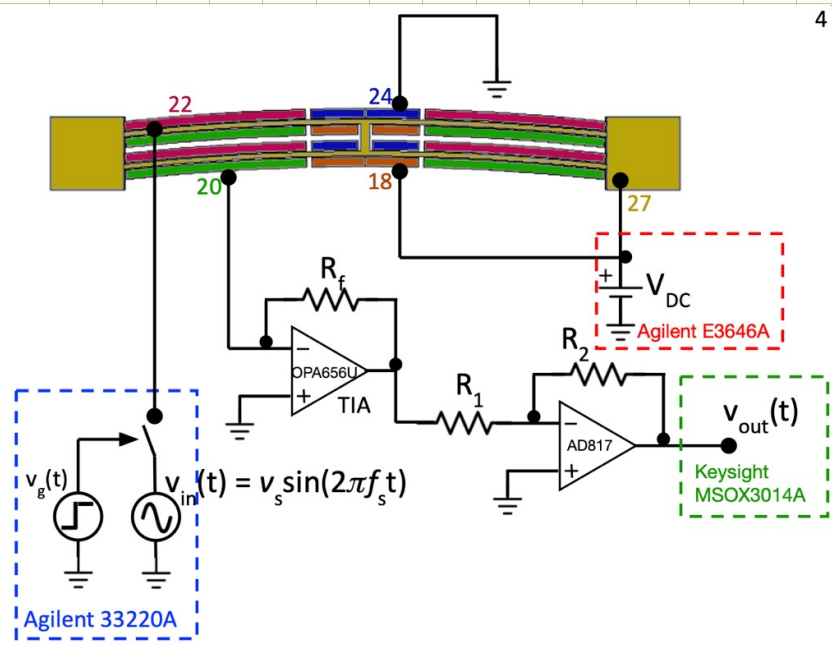
• Zero Crossing

oppure "gradino di forza
statico"?



spigoli → Fourier
= Alte ~~trans~~ ~~trans~~

Ringium = TIA che oscilla
sottosmorzato



$$1p : q_2 = 0$$

$$V_T = 0$$

eccitiamo solo il primo modo

↳ $V_{in}(t)$, V_{DC} relativamente bassi

$$F_i(q) = \cancel{\dot{\phi}_{DD}(q)} V_{DC}^2 + \cancel{\dot{\phi}_{TT}(q)} V_T^2 + 2\cancel{\dot{\phi}_{DT}(q)} V_{DC} V_T + 2\dot{\phi}_{DA}(q) V_{DC} V_{AC} \sin(\omega t)$$

Nota: anche se $V_T = 0$

gli elettrodi di tuning restano nella geometria

↳ il loro effetto resta incluso in $\dot{\phi}_{DD}$ $\dot{\phi}_{DA}$

ROM del PAPER

$$M=1$$

$$M \ddot{q}_i + M B_i(q) = F_i(q)$$

$$\ddot{q}_i + B_i(q) = F_i(q)$$

$$q_2 = 0 \quad B_1(q_1, 0) = B_1(q_1) \approx \cancel{C_0^1} + C_1^1 q_1$$

$$C_0^1 = -9,72 \times 10^{-9} \text{ mm/ms}^2 \quad \text{per adesso trascurare}$$

No forze esterne

se $F_i(q) = 0$

$$\ddot{q}_1 + C_1^1 q_1 = 0$$

Oscillatore armonico

$$\omega_{01}^2 = C_1^1 \Rightarrow \omega_{01} = \sqrt{C_1^1} = 2,62 \frac{1}{\text{ms}}$$

$$C_1^1 = 6,85 \text{ ms}^{-2}$$

$$f_{01} = \frac{\omega_{01}}{2\pi} \approx 416 \text{ kHz}$$